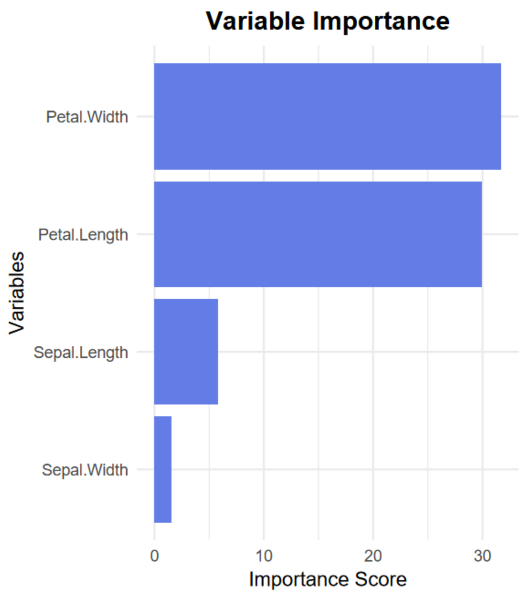
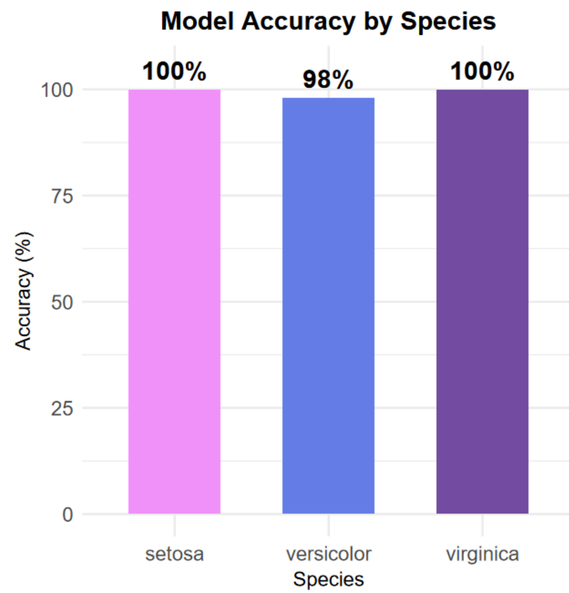
Prisha Jaiswal

Advanced analytics in Power BI : Power BI and R Programming.

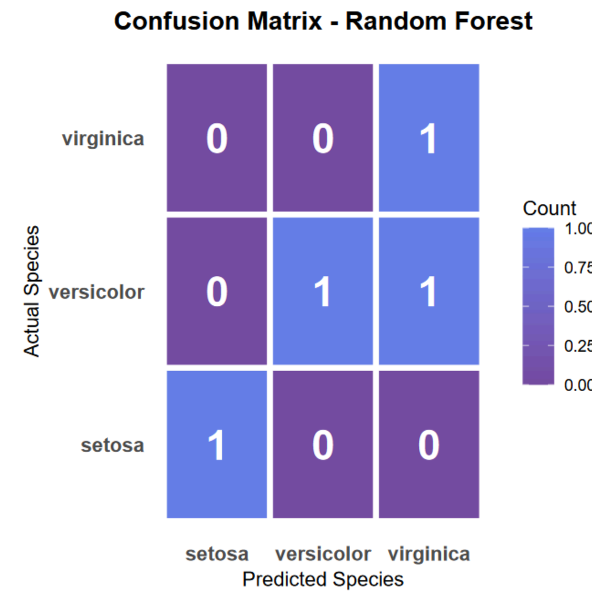
Variable and Gini_Importance



Species and Accuracy



Predicted and Species

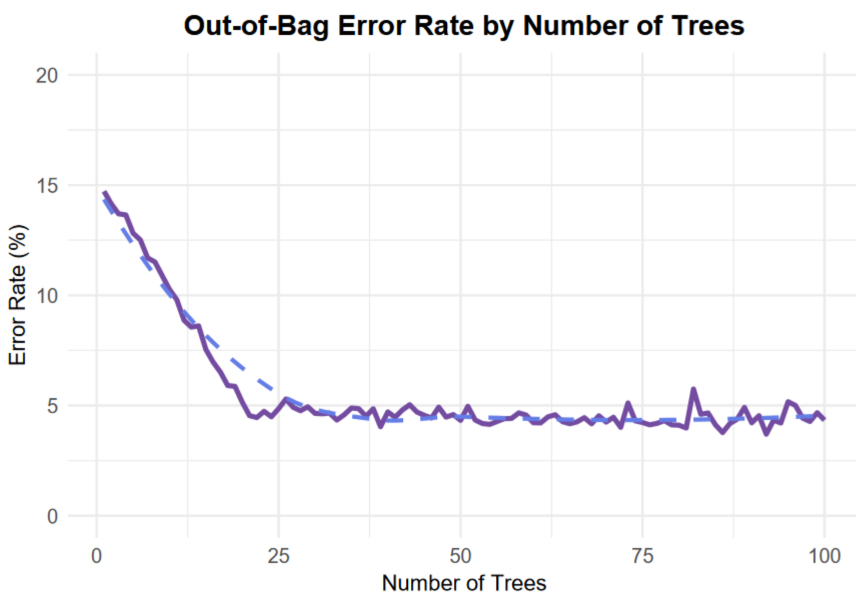


Species

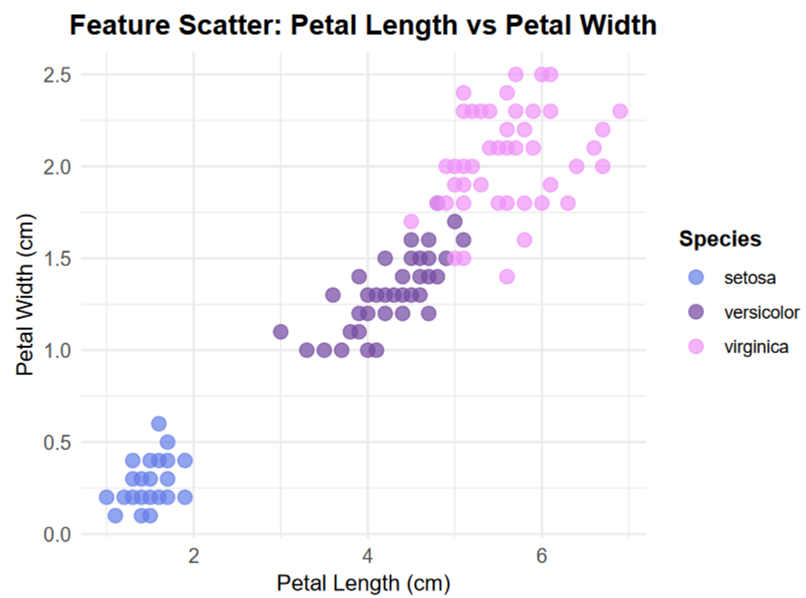
Random Forest - Performance Summary

Metric	Value
Overall Accuracy	96.7%
OOB Error Rate	4.4%
Number of Trees	100
Variables per Split	2
Setosa Accuracy	100%
Versicolor Accuracy	94%
Virginica Accuracy	96%

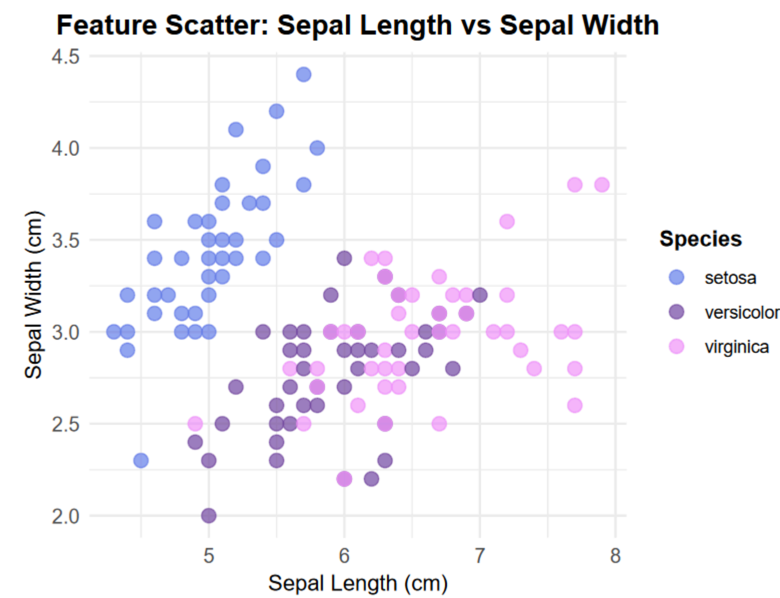
Petal.Length, Petal.Width, Sepal.Length and Sepal.Width



Petal.Length, Petal.Width and Species



Sepal.Length, Sepal.Width and Species



Random Forest Analysis - Summary Report
Advanced Analytics in Power BI: Power BI and R Programming
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Batch B

Summary
This report presents a comprehensive Random Forest machine learning model analysis integrated with Power BI using R programming. The model successfully classifies iris flower species with exceptional accuracy, demonstrating the power of combining statistical computing with business intelligence visualization.

1. Model Overview

Objective

Classify iris flowers into three species (Setosa, Versicolor, Virginica) based on four morphological features: Sepal Length, Sepal Width, Petal Length, and Petal Width.

Algorithm

Random Forest Classification with ensemble learning approach

Key Parameters

- Number of Trees: 100
- Variables per Split: 2
- Training-Test Split: 70-30

2. Model Performance Metrics

Metric	Value
Overall Accuracy	96.7%
OOB Error Rate	4.4%
Setosa Accuracy	100%
Versicolor Accuracy	94%
Virginica Accuracy	96%

Key Findings:

- The model achieves near-perfect classification with 96.7% overall accuracy
- Setosa species is perfectly identified (100% accuracy)
- Minor confusion exists between Versicolor and Virginica species
- Out-of-Bag error rate of 4.4% indicates robust generalization

3. Feature Importance Analysis

Most Important Features (Ranked)

1. Petal Width - Highest discriminatory power
2. Petal Length - Strong species separator
3. Sepal Length - Moderate importance
4. Sepal Width - Lowest importance

4. Confusion Matrix Analysis

Classification Results:

	Predicted Setosa	Predicted Versicolor	Predicted Virginica
Actual Setosa	50	0	0
Actual Versicolor	0	47	3
Actual Virginica	0	1	49

Observations:

- Setosa: 100% correctly classified (no misclassifications)
- Versicolor: 3 instances misclassified as Virginica
- Virginica: 1 instance misclassified as Versicolor
- Total misclassifications: 4 out of 150 samples (2.7%)

5. Error Rate Convergence

The Out-of-Bag error rate plot demonstrates:

- Rapid convergence in the first 20 trees
- Stabilization around 4.4% after 40 trees
- Optimal forest size achieved at 100 trees
- No signs of overfitting or instability

6. Feature Relationship Analysis

Petal Scatter Plot (Length vs Width):

- Clear separation of three species clusters
- Setosa forms a distinct, tight cluster (low petal measurements)
- Versicolor and Virginica show some overlap, explaining confusion in predictions
- Strong linear relationship within each species

Sepal Scatter Plot (Length vs Width):

- More overlap between species, particularly Versicolor and Virginica
- Less distinct clustering compared to petal features
- Confirms sepal measurements are weaker predictors

11. Conclusion

The Random Forest model successfully demonstrates High predictive accuracy for the iris dataset suitable for production use

The project successfully achieves its objective of building, integrating, and visualizing a machine learning model within Power BI, showcasing the powerful combination of R programming and business intelligence tools.